WEEK 5 / LESSON 6 / 50 MINUTES

L6

LoRA · PEFT · Domain Adaptation

90% of "we need to fine-tune" conversations should have ended with "RAG + better prompts". But when FT is right — strict format, specialised jargon, latency — LoRA makes it cheap enough that a student laptop can run it.

Has anyone in class ever fine-tuned a model? If no, you're in good company.

WHAT'S INSIDE

WHY THIS LESSON MATTERS

Fine-tune rarely. When you do — adapt small, ship fast.

WHY
Real
A real reason this matters in industry today.

SCALE
Today
A real reason this matters in industry today.

SPEED
Now
A real reason this matters in industry today.

IMPACT
Yours
A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Full FT vs PEFT vs LoRA.
2. Why LoRA works — the intuition.
3. Prepare chat-JSONL dataset.
4. Fine-tune 7B model on consumer GPU.
5. Fine-tuned vs base.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
When to fine-tune (rarely)
The decision ladder

02
Full FT vs PEFT vs LoRA
Key concept of this lesson.

03
Why LoRA works
Low-rank approximation of weight update

04
Data is the lever
500 high-quality > 50,000 noisy

01

PART 1 OF 4

When to fine-tune (rarely)

The decision ladder

CONCEPT 1 OF 4

When to fine-tune (rarely)

1. Try RAG.
2. Try better prompts.
3. Try few-shot.
4. ONLY THEN: fine-tune.

RULE

4th option

Fine-tuning is rarely the right first answer.

DEFINITION

When to fine-tune (rarely)

The decision ladder

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Full FT vs PEFT vs LoRA

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

CONCEPT 3 OF 4

Why LoRA works

Frozen W (base weights).
Trainable $\Delta W = B \cdot A$, $\text{rank}(\Delta W) \ll \text{rank}(W)$.
Effective weight at inference: $W + B \cdot A$.
Merge at deploy; zero runtime overhead.
INSIGHT
100–1000× fewer params

Surprisingly enough to adapt domain.

DEFINITION

Why LoRA works

Low-rank approximation of weight update

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Data is the lever

Every example uses target output format.
Cover edge cases.
Include refusals for out-of-scope.
Human-review before training.
QUALITY
Beats quantity

Every time. Always.

KEY POINTS

- Every example uses target output format
- Cover edge cases
- Include refusals for out-of-scope
- Human-review before training

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Run PEFT code on T4

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1 *Is there a production problem where fine-tuning would genuinely help?*

Q2 *What's your minimum viable dataset?*

Q3 *How would you roll back if the fine-tune regressed?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Fine-tune when:

A Always

B Never

C RAG + prompts fail AND format/latency constrained

D Just for fun

ANSWER · QUESTION 1 OF 3

Fine-tune when:

C

CORRECT ANSWER

RAG + prompts fail AND format/latency constrained

WHY

Decision ladder — RAG + prompts first

KNOWLEDGE CHECK · QUESTION 2 OF 3

LoRA trains:

A 100%

B ~50%

C < 1%

D All decoder

ANSWER · QUESTION 2 OF 3

LoRA trains:

CORRECT ANSWER

C

< 1%

WHY

< 0.5% at rank=8 — small low-rank matrices

KNOWLEDGE CHECK · QUESTION 3 OF 3

QLoRA adds:

A Quantisation

B Quorum

C Queues

D Questions

ANSWER · QUESTION 3 OF 3

QLoRA adds:

CORRECT ANSWER

A

Quantisation

WHY

4-bit quantisation → fits 8B model on 6 GB GPU

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

1. Try RAG.
2. Try better prompts.
3. Try few-shot.
4. ONLY THEN: fine-tune.

RULE
4th option

Fine-tuning is rarely the right first answer.

THE TAKEAWAY

"Adapt small; ship fast."

WHAT TO NOTICE

1. When to fine-tune (rarely)
2. Full FT vs PEFT vs LoRA
3. Why LoRA works
4. Data is the lever

APPLY IT → ASSIGNMENT

Mini assignment — your first LoRA

Fine-tune Llama-3-8B on 200 examples in your domain.

DELIVERABLES

- Prepare JSONL.
- QLoRA on Colab T4.
- Evaluate vs base.
- Post before/after.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Repo + eval numbers.

ONE THING TO REMEMBER

"Adapt small; ship fast."

— AI Learning Hub

END OF LESSON 5

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 7 — Evaluation + Guardrails.